S.No: 1	Exp. Name: ***Quick Sort***	Date:2026-07-13

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Aim:

Write a program to sort an array of length n using the Quick Sort algorithm. The program should read the array elements and print the array before and after sorting.

Input Format:

- The first line contains a single integer n, representing the number of elements in the array.
- The second line contains n space-separated integers representing the elements of the array.

Output Format:

- The first line of output contains space-separated integers representing the elements of the original array
- The Second input line contains space-separated integers representing the elements after sorting.

Constraints:

- $1 \leq n \leq 1000$

Note:

- The driver code is already provided. You need to complete the required code to perform Quick Sort correctly.

Source Code:

Quick_sort.c

```
#include <stdio.h>
int partition(int arr[], int low, int high) {
    int pivot = arr[high];
    int i = low - 1;

    for (int j = low; j < high; j++) {
        if (arr[j] <= pivot) {
            i++;
            int temp = arr[i];
            arr[i] = arr[j];
            arr[j] = temp;
        }
    }

    int temp = arr[i + 1];
    arr[i + 1] = arr[high];
    arr[high] = temp;

    return i + 1;
}

// Quick Sort function
void quickSort(int arr[], int low, int high) {
    // Complete the code...
    if (low < high) {
        int pi = partition(arr, low, high);
```

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```
        quickSort(arr, low, pi - 1);
        quickSort(arr, pi + 1, high);
        }


}

// Function to print an array
void printArray(int arr[], int size) {
    for (int i = 0; i < size; i++) {
        printf("%d ", arr[i]);
    }
    printf("\n");
}

int main() {
    int n;
    // Input the size of the array
    scanf("%d", &n);
    int arr[n];

    // Input the elements of the array
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    // Print the original array
    printArray(arr, n);

    // Sort the array using quick sort
    quickSort(arr, 0, n - 1);

    // Print the sorted array
    printArray(arr, n);

    return 0;
}
```

Execution Results - All test cases have succeeded!

Test Case - 1
User Output
4
1 8 3 4
1 8 3 4
1 3 4 8

Test Case - 2
User Output
7
2 9 -7 0 2 -4 1
2 9 -7 0 2 -4 1

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-7 -4 0 1 2 2 9